

The Good, the Grey and the Redundant - Data Redundancy and Grey Literature at the EDC

Oliver Brough, Peter Holt, and Vivek Mistry - UKRI/STFC

The UK Energy Research Centre (www.ukerc.ac.uk) undertakes “Independent whole systems research for a sustainable energy future” and was formed in 2004. The Energy Data Centre (EDC) is a capability of UKERC and provides a discovery portal and data management expertise for researchers funded through UKERC.

Current key priorities for the EDC are:



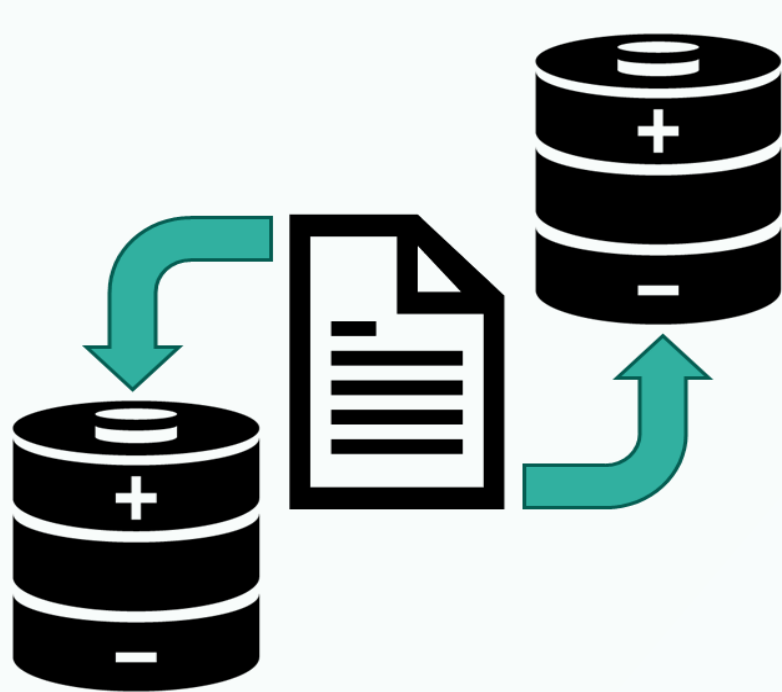
The Energy Data Centre

- The EDC is a digital repository for UK focused energy-related data and content with the goal to unify both the technical and social science aspects of energy research, and has been running continuously for 25 years without a major outage
- The main content the EDC stores are energy related datasets, publications, and project metadata and as of Aug 2026, the EDC holds 365 datasets, 1847 publications, and information on 8853 projects



Grey Literature and ETI

- One of the key focuses of the EDC’s preservation remit revolves around Grey Literature, which are documents that have not been formally published and are usually only available on a specific domain/website
- In the Energy sector, project-driven research initiatives often host their research outputs on dedicated project websites, leading to a potential loss of data once these websites are terminated at the end of the project’s life
- For example, the Energy Technologies Institute (ETI) was a private-public partnership which ran from 2007-2019 and produced many reports that were only available from the ETI website, meaning that once the project’s website was to be terminated, these outputs would be unavailable to the public
- Before the website was shut down, the EDC were given the rights to host ETI’s data and reports, with these copies on the EDC becoming the only way to publicly access these files
- This means that if the files on the EDC end up being corrupted, this information could be permanently lost



Data Redundancy

- To avoid this potential catastrophe with grey literature, the EDC practices Data Redundancy
- This is where additional copies of a digital object are stored across multiple locations, ensuring that if there is any damage to one storage location, the data is not lost and can be retrieved from the other locations
- The EDC stores most* of its content across 4 servers in 2 locations, Rutherford Appleton Laboratory and Daresbury Lab. This has allowed the EDC to avoid major outages and minimise disruptions to users during periods of server instability, and has bolstered our preservation practices by maintaining multiple copies of our content in case of a loss of file integrity
- * The EDC’s Datasets are stored with the Centre for Environmental Data Analysis (CEDA) who have their own systems for data management

The Integrity Checker

- Validating the integrity of files following data redundancy is a crucial part of preservation practice
- For the grey literature publications the EDC holds, the team developed an in house-integrity checker tool to ensure proper preservation of these files across all of our servers
- The tool conducts a checksums validation on the 4 main EDC servers, and across our preservation copy in CEDA against our master checksum recorded at ingest
- Having this tool has given us an easy way to validate the integrity of our files and investigate any potential issues
- This integrity check also has functionality outside of our stored grey literature, as it conducts link validation to ensure that any links on our service still have valid landing points
- The results of these checks are in a csv format that can be combined into a single spreadsheet containing all fails to be reviewed
- This tool was developed in house due to our practice of data redundancy making it difficult for pre-existing tools to able to access all the necessary servers to validate file integrity
- This tool serves as part of the EDCs continued development in its preservation practices

What’s Next?

- The EDC will be moving towards a cloud-based storage system, so the tool will be updated to cover the shifting data redundancy structure of the EDC
- The tool will continue to be improved and will be further established within our preservation workflow to ensure resilient preservation of grey literature

UKEDC Integrity Checker

Environment Variables set:

Show environment variables

Run Checks:

Enter custom file name (optional):

Run Checksum Checker

Run Database Link Checker

Run DOI Link Checker

Run Automated Link Checker

Logs:

[2026-08-11 10:59:11] CHECKSUM CHECKER: Successfully created checksum comparison output at 2026-08-11_10-45-26-checksum_checker.csv, Successful: 0, Unsuccessful: 1833

Checksum for One File:

Enter file UUID:

Get Checksums

Download results

CHECKSUM_CHECKER_RESULTS/2026-08-11_10-45-26-checksum_checker.csv	11-08-26 10:59:11	Download
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